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HostelSense

Smart Hostel Fee & Room Report System

A C++ Object Oriented Programming Project

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1 Aim

To write a C++ program that works out the hostel fee for a room according to its type, splits the extra electricity bill of an AC room among the students living in it, and saves the final report to a text file. The program is to be written using classes and inheritance, and it should reject repeated roll numbers and wrong inputs.

2 Introduction

Hostel fees are not the same for every room. A room with an attached bathroom costs more than one without, and an AC room costs more again. On top of that, AC rooms use a lot of electricity, and that bill is not part of the fixed fee. If the hostel office does all this by hand for every room, small mistakes are easy to make: a wrong amount for a room type, a student entered twice, or an electricity bill divided wrongly.

HostelSense is a small console program that handles this for one room at a time. The user picks the room type, enters the room code and the details of the three students living in it, and the program prints a report on the screen and also stores it in a file called `HostelReport.txt`.

The project was made as a practical for the Object Oriented Programming course. The idea was to take a problem that is easy to understand, and use it to practise the main parts of the course in one place: classes, inheritance, arrays, loops, decisions and file handling.

2.1 Problem Statement

- **Manual calculation.** Fees depend on room type, so working them out by hand invites mistakes.
- **Duplicate entries.** Nothing stops the same student from being entered twice.
- **Unclear bills.** The shared AC electricity cost is hard to split, and it gets mixed up with the hostel fee.

2.2 Objectives

1. Calculate the hostel fee automatically from the room type.
2. Block the same roll number from being registered twice in a room.

3. Split the AC electricity bill equally and keep it separate from the hostel fee.
4. Save the report in a text file as well as showing it on screen.
5. Use inheritance so that AC-specific details are added without rewriting the common code.

2.3 Scope

The program is meant for a hostel office that deals with rooms of three students each. It deals with one room per run and produces one report. It is a console program, so there is no graphical interface and no database. The scope is kept small on purpose, so that the OOP ideas stay clear in the code.

3 System Requirements

3.1 Software

Item	Details
Language	C++
Compiler / IDE	Online GDB (g++ compiler)
Operating system	Any (Windows, Linux or macOS, since the program uses only standard C++)
Libraries	`iostream`, `fstream`, `string`

Table 1: Software requirements

3.2 Hardware

The program is very light. Any computer that can run a browser (for Online GDB) or a C++ compiler is enough. No special memory, graphics or storage is needed, apart from a few bytes of disk space for the report file.

3.3 Header files used

Header	Why it is needed
`iostream`	`cin` and `cout` for input and output on the console
`fstream`	`ofstream` for writing the report to `HostelReport.txt`
`string`	the `string` type for names, room codes and roll numbers

Table 2: Header files

The statement `using namespace std;` is used so that `cin`, `cout`, `string` and `ofstream` can be written without the `std::` prefix.

4 Features

- **Three room types.** The fee per student is fixed for each type, as shown in Table 3.
- **Unique roll numbers.** A college roll number cannot repeat inside a room.
- **Separate electricity bill.** For AC rooms the bill is units × rate, divided equally among the three students. It is never added to the hostel fee.
- **Saved report.** The same report is printed on screen and written to `HostelReport.txt`.
- **Basic input checks.** An invalid room choice, or negative units or rate, is rejected.

Choice	Room type	Fee per student (Rs.)
1	Attached bathroom, Non-AC	95,000
2	Non-attached bathroom, Non-AC	73,000
3	AC with attached bathroom	1,25,000

Table 3: Room types and hostel fee

4.1 How the amounts are worked out

For a room with three students, the amounts are found as follows.

$$\begin{aligned} \text{Room fee total} &= 3 \times \text{fee per student} \\ \text{Electricity bill} &= \text{units consumed} \times \text{rate per unit} \\ \text{Share of each student} &= \text{Electricity bill} \div 3 \end{aligned}$$

The electricity bill is only asked for in AC rooms. For the other two types it is taken as zero.

5 Design of the Program

5.1 Algorithm

1. Start.

2. Display the three room types with their fees and read the choice.
3. If the input is not a number, or the choice is not 1, 2 or 3, print "Invalid choice!" and stop.
4. If the choice is 3 (AC room):
   (a) read the room code, and the name and roll number of three students (checking each roll number against the earlier ones);
   (b) read the units consumed and the rate per unit;
   (c) if either is negative or unreadable, print "Invalid electricity details!" and stop;
   (d) set the fee to 1,25,000 and the bill to units × rate.
5. Otherwise (choice 1 or 2):
   (a) read the room code, names and roll numbers in the same way;
   (b) set the fee to 95,000 for choice 1 or 73,000 for choice 2, and the bill to 0.
6. Open `HostelReport.txt` for writing. If it cannot be opened, print "Cannot create report!" and stop.
7. Print the report on the screen and write the same report to the file.
8. Close the file, print "Report saved successfully!".
9. Stop.

5.2 Flowchart

Figure 1 shows the same steps as a flowchart.

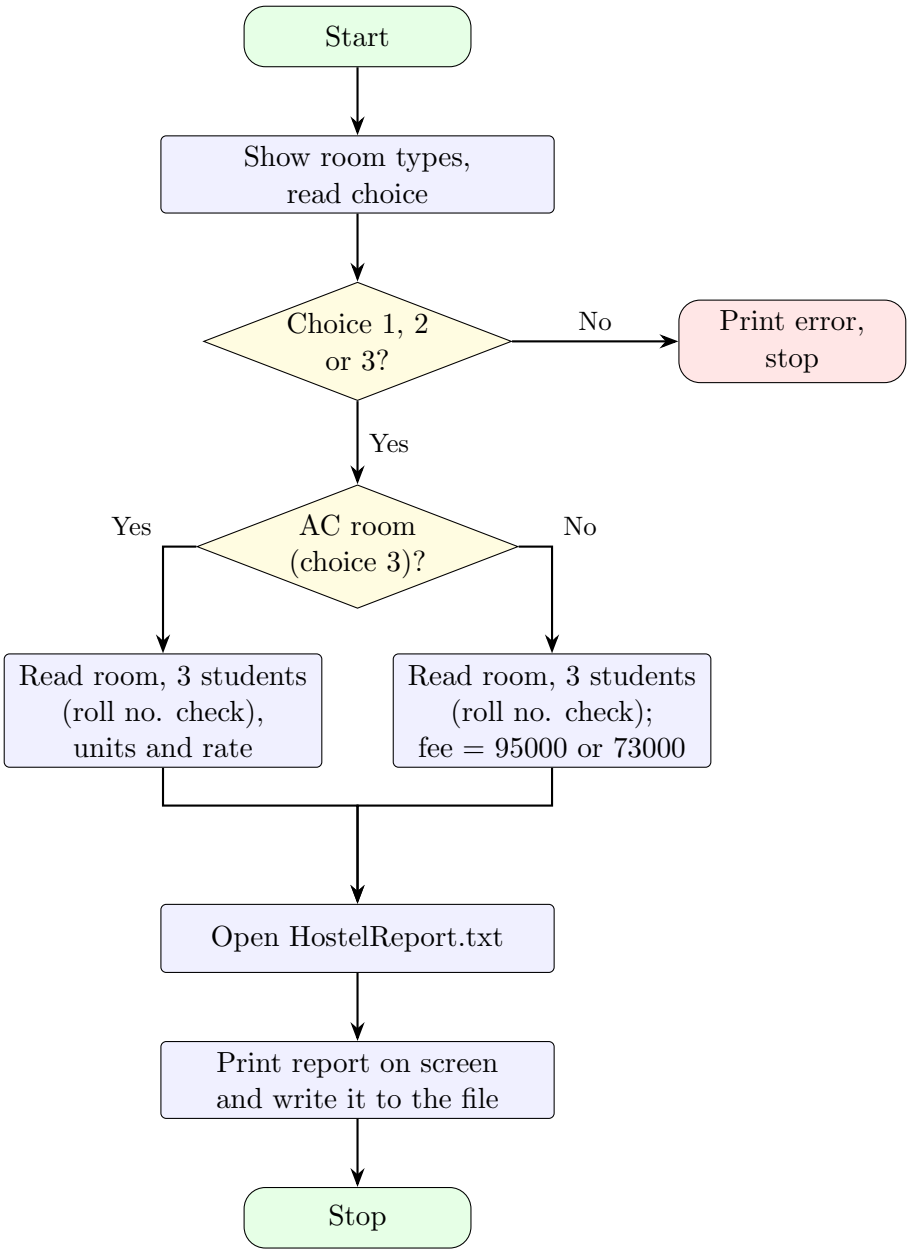


Figure 1: Flowchart of HostelSense

5.3 Classes

The program has two classes.

Room is the base class. It holds the room code and two arrays of three elements each, one for the names and one for the roll numbers. It has two functions:

- `input()` reads the room code, names and roll numbers, and does the duplicate roll number check.
- `show()` prints the full report. It takes the output stream, the room type, the electricity bill and the fee as arguments.

ACRoom is derived from `Room` using public inheritance. It adds the data members `units` and `rate`, a constructor that sets both to zero, `inputAC()` to read them, and `bill()` which returns `units * rate`.

Since `ACRoom` inherits from `Room`, an AC room object can call `input()` and `show()` directly without rewriting them. Only the extra electricity part had to be added.

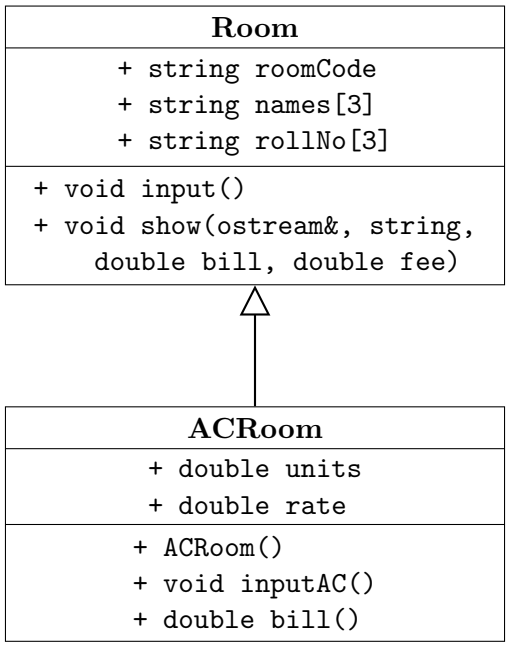


Figure 2: Class diagram: `ACRoom` inherits from `Room`

Class	Data members	Member functions
`Room`	`roomCode`, `names[3]`, `rollNo[3]`	`input()`, `show()`
`ACRoom`	inherits the above, adds `units`, `rate`	`ACRoom()`, `inputAC()`, `bill()`

Table 4: Summary of classes

5.4 OOP concepts used

Class and objects `Room` stores everything about one room. `main()` creates one object of `Room` and one of `ACRoom`, and uses whichever one the room type needs.

Inheritance `ACRoom` extends `Room` and adds the electricity details.

Encapsulation The data of a room and the functions that work on it are kept together in the class.

Constructor `ACRoom()` sets `units` and `rate` to zero, so they never hold garbage values if they are used before being read.

Function reuse `show()` takes an `ostream&`, so the same function prints to the screen (`cout`) and to the file (`ofstream`). The report code is written only once.

File handling An `ofstream` object creates and writes `HostelReport.txt`.

5.5 Why a stream reference is passed to `show()`

An `ofstream` and `cout` are both kinds of output stream. Because `show()` takes the parameter as `ostream &out`, either of them can be passed in. If this was not done, the report would have to be written twice, once with `cout << ...` and once with `f << ...`, and any change in the format would have to be made in both places. With one function, the screen and the file can never get out of step.

6 Implementation

The complete source code is given in the Appendix. This section goes through it function by function.

6.1 The `Room::input()` function

This function reads the room code first. Then it runs a `for` loop three times, once for each student. Inside the loop it reads the name and then the roll number.

For the name, the program uses `getline(cin >> ws, names[i])`. A student's name can contain spaces, so a plain `cin >> name` would read only the first word. The `ws` manipulator removes the leftover newline from the previous input, so `getline` does not read an empty line by mistake.

6.1.1 Checking for duplicate roll numbers

Each roll number is read inside a `while (true)` loop. After reading it, the program compares it with the roll numbers of all the earlier students in the same room. If a match is found, a message is printed and the loop runs again. If there is no match, the loop breaks and the next student is taken.

```
1 while (true) {
2     cout << "Enter student " << i + 1 << " college roll no: ";
3     cin >> rollNo[i];
4
```

```
    bool duplicate = false;
    for (int j = 0; j < i; j++) {
        if (rollNo[j] == rollNo[i])
            duplicate = true;
    }

    if (duplicate)
        cout << "This roll no already lives in the hostel!
           Enter again.\n";
    else
        break;
}
```

The inner loop runs from `0` to `i-1`, so the first student has nothing to be compared with, the second is compared with one student, and the third with two.

6.2 The `Room::show()` function

`show()` builds the whole report. It first works out the share of each student as `bill / 3`. Then it sets the stream to fixed-point format with two decimal places:

```
out.setf(ios::fixed);
out.precision(2);
```

This is why `125000` appears as `125000.00` in the report. After that, the report is printed in three parts: the heading with the room code and type, the hostel fee for each student with the room total, and the electricity bill with each student's share.

6.3 The `ACRoom` class

`inputAC()` reads the units consumed and the rate per unit. `bill()` multiplies them. These are the only extra things an AC room needs.

```
double bill() {
    return units * rate;
}
```

For non-AC rooms the bill passed to `show()` is 0, so every share comes out as Rs. 0.00 and the same function can be used for all room types.

6.4 The `main()` function

`main()` does the following, in order:

1. prints the menu and reads the choice;
2. rejects the choice if `cin.fail()` is true or the number is outside 1 to 3;
3. creates a `Room` object `r` and an `ACRoom` object `c`;
4. for choice 3, uses `c` to read the room and the electricity details, checks them, and sets the fee to 125000 and the bill to `c.bill()`;
5. for choices 1 and 2, uses `r` to read the room and sets the fee to 95000 or 73000;
6. opens the report file and prints the report on screen and in the file.

6.5 Printing to the screen and the file together

Because `show()` accepts any output stream, `main()` calls it twice with the same data, once with `cout` and once with the file object.

```
ofstream f("HostelReport.txt");
if (!f) {
    cout << "Cannot create report!\n";
    return 0;
}

if (choice == 3) {
    c.show(cout, type, bill, fee);
    c.show(f, type, bill, fee);
}
else {
    r.show(cout, type, bill, fee);
    r.show(f, type, bill, fee);
}
f.close();
```

6.6 Input validation

There are three checks in `main()`:

- the room type must be 1, 2 or 3 (and a non-number such as a letter is caught by `cin.fail()`);
- for AC rooms, units and rate must not be negative;
- if the report file cannot be created, the program says so and exits.

6.7 Variables used in `main()`

Variable	Type	Purpose
`choice`	`int`	room type selected by the user
`r`	`Room`	object used for non-AC rooms
`c`	`ACRoom`	object used for AC rooms
`type`	`string`	room type text printed in the report
`bill`	`double`	electricity bill of the room
`fee`	`double`	hostel fee per student
`f`	`ofstream`	the report file

Table 5: Main variables in `main()`

7 Output

This section shows the program running. The names in the test data are placeholders (A to R), and the roll numbers are only samples. The first two runs are the ones from the screenshots. The remaining runs were done to check the other room type and the error cases.

7.1 Screenshots of the report

7.1.1 AC room (3S12)

Room type 3 was selected. The students were D, E and F, and the electricity bill came to Rs. 300.

```
--- HostelSense Report ---
Room: 3S12
Type: AC with Attached Bathroom

=== HOSTEL FEE ===
Student 1: D (Roll no: 25PE307)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Student 2: E (Roll no: 24ME300)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Student 3: F (Roll no: 25EL402)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Total hostel fee from this room: Rs. 375000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 300.00
D (Roll no: 25PE307) pays: Rs. 100.00
E (Roll no: 24ME300) pays: Rs. 100.00
F (Roll no: 25EL402) pays: Rs. 100.00
```

Figure 3: Report for AC room 3S12

The fee of Rs. 1,25,000 is charged to each of the three students, which gives Rs. 3,75,000 for the room. The electricity bill of Rs. 300 is shown separately and each student pays Rs. 100.

7.1.2 Non-AC room with attached bathroom (4S13)

Room type 1 was selected. The students were A, B and C.

```
--- HostelSense Report ---
Room: 4S13
Type: Attached Bathroom, Non-AC

=== HOSTEL FEE ===
Student 1: A (Roll no: 25EC407)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Student 2: B (Roll no: 25ME013)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Student 3: C (Roll no: 25EL607)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Total hostel fee from this room: Rs. 285000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 0.00
A (Roll no: 25EC407) pays: Rs. 0.00
B (Roll no: 25ME013) pays: Rs. 0.00
C (Roll no: 25EL607) pays: Rs. 0.00
```

Figure 4: Report for non-AC room 4S13

The fee is Rs. 95,000 per student, so Rs. 2,85,000 for the room. There is no electricity bill for a non-AC room, so each student pays Rs. 0.00 under that heading.

7.2 Complete console sessions

The listings below show everything that appears on the console, from the menu to the last message, including what was typed in.

7.2.1 Run 1: AC room, 3S12

```
1. Attached bathroom, Non-AC     (Rs. 95,000)
2. Non-attached bathroom, Non-AC (Rs. 73,000)
3. AC with attached bathroom     (Rs. 1,25,000)
Select room type: 3
Enter room code (e.g. 4S13): 3S12
Enter student 1 name: D
Enter student 1 college roll no: 25PE307
Enter student 2 name: E
Enter student 2 college roll no: 24ME300
Enter student 3 name: F
Enter student 3 college roll no: 25EL402
Enter AC units consumed: 300
Enter rate per unit (Rs.): 1

--- HostelSense Report ---
Room: 3S12
Type: AC with Attached Bathroom

=== HOSTEL FEE ===
Student 1: D (Roll no: 25PE307)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Student 2: E (Roll no: 24ME300)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Student 3: F (Roll no: 25EL402)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Total hostel fee from this room: Rs. 375000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 300.00
D (Roll no: 25PE307) pays: Rs. 100.00
E (Roll no: 24ME300) pays: Rs. 100.00
F (Roll no: 25EL402) pays: Rs. 100.00
Report saved successfully!
```

7.2.2 Run 2: Non-AC room with attached bathroom, 4S13

```
1. Attached bathroom, Non-AC     (Rs. 95,000)
2. Non-attached bathroom, Non-AC (Rs. 73,000)
3. AC with attached bathroom     (Rs. 1,25,000)
Select room type: 1
Enter room code (e.g. 4S13): 4S13
Enter student 1 name: A
Enter student 1 college roll no: 25EC407
Enter student 2 name: B
Enter student 2 college roll no: 25ME013
Enter student 3 name: C
Enter student 3 college roll no: 25EL607

--- HostelSense Report ---
Room: 4S13
Type: Attached Bathroom, Non-AC

=== HOSTEL FEE ===
Student 1: A (Roll no: 25EC407)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Student 2: B (Roll no: 25ME013)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Student 3: C (Roll no: 25EL607)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Total hostel fee from this room: Rs. 285000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 0.00
A (Roll no: 25EC407) pays: Rs. 0.00
B (Roll no: 25ME013) pays: Rs. 0.00
C (Roll no: 25EL607) pays: Rs. 0.00
Report saved successfully!
```

7.2.3 Run 3: Non-AC room without attached bathroom, 2S05

```
1. Attached bathroom, Non-AC     (Rs. 95,000)
2. Non-attached bathroom, Non-AC (Rs. 73,000)
3. AC with attached bathroom     (Rs. 1,25,000)
Select room type: 2
Enter room code (e.g. 4S13): 2S05
Enter student 1 name: G
Enter student 1 college roll no: 25CE101
Enter student 2 name: H
Enter student 2 college roll no: 25CE102
Enter student 3 name: I
Enter student 3 college roll no: 25CE103
```

```
--- HostelSense Report ---
Room: 2S05
Type: Non-Attached Bathroom, Non-AC

=== HOSTEL FEE ===
Student 1: G (Roll no: 25CE101)
   TOTAL HOSTEL FEE PAYABLE : Rs. 73000.00
Student 2: H (Roll no: 25CE102)
   TOTAL HOSTEL FEE PAYABLE : Rs. 73000.00
Student 3: I (Roll no: 25CE103)
   TOTAL HOSTEL FEE PAYABLE : Rs. 73000.00
Total hostel fee from this room: Rs. 219000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 0.00
G (Roll no: 25CE101) pays: Rs. 0.00
H (Roll no: 25CE102) pays: Rs. 0.00
I (Roll no: 25CE103) pays: Rs. 0.00
Report saved successfully!
```

The fee here is Rs. 73,000 per student, which gives Rs. 2,19,000 for the room.

7.2.4 Run 4: Duplicate roll number

Student K was first entered with the same roll number as student J. The program refused it and asked again. After a different roll number was given, it carried on normally.

```
1. Attached bathroom, Non-AC     (Rs. 95,000)
2. Non-attached bathroom, Non-AC (Rs. 73,000)
3. AC with attached bathroom     (Rs. 1,25,000)
Select room type: 1
Enter room code (e.g. 4S13): 1S01
Enter student 1 name: J
Enter student 1 college roll no: 25IT201
Enter student 2 name: K
Enter student 2 college roll no: 25IT201
This roll no already lives in the hostel! Enter again.
Enter student 2 college roll no: 25IT202
Enter student 3 name: L
Enter student 3 college roll no: 25IT203

--- HostelSense Report ---
Room: 1S01
Type: Attached Bathroom, Non-AC

=== HOSTEL FEE ===
```

```
Student 1: J (Roll no: 25IT201)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Student 2: K (Roll no: 25IT202)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Student 3: L (Roll no: 25IT203)
   TOTAL HOSTEL FEE PAYABLE : Rs. 95000.00
Total hostel fee from this room: Rs. 285000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 0.00
J (Roll no: 25IT201) pays: Rs. 0.00
K (Roll no: 25IT202) pays: Rs. 0.00
L (Roll no: 25IT203) pays: Rs. 0.00
Report saved successfully!
```

7.2.5 Run 5: Invalid room type

```
1. Attached bathroom, Non-AC     (Rs. 95,000)
2. Non-attached bathroom, Non-AC (Rs. 73,000)
3. AC with attached bathroom     (Rs. 1,25,000)
Select room type: 7
Invalid choice!
```

7.2.6 Run 6: Negative rate for electricity

```
1. Attached bathroom, Non-AC     (Rs. 95,000)
2. Non-attached bathroom, Non-AC (Rs. 73,000)
3. AC with attached bathroom     (Rs. 1,25,000)
Select room type: 3
Enter room code (e.g. 4S13): 3S09
Enter student 1 name: M
Enter student 1 college roll no: 25CH11
Enter student 2 name: N
Enter student 2 college roll no: 25CH12
Enter student 3 name: O
Enter student 3 college roll no: 25CH13
Enter AC units consumed: 100
Enter rate per unit (Rs.): -5
Invalid electricity details!
```

In this case the program stops without creating a report.

7.2.7 Run 7: AC room with a decimal rate, 3S10

This run uses 250 units at Rs. 8.50 per unit. The bill is $250 \times 8.5 = 2125$, and the share is $2125 \div 3 = 708.33$ (rounded to two decimal places).

```
1. Attached bathroom, Non-AC     (Rs. 95,000)
2. Non-attached bathroom, Non-AC (Rs. 73,000)
3. AC with attached bathroom     (Rs. 1,25,000)
Select room type: 3
Enter room code (e.g. 4S13): 3S10
Enter student 1 name: P
Enter student 1 college roll no: 25EC11
Enter student 2 name: Q
Enter student 2 college roll no: 25EC12
Enter student 3 name: R
Enter student 3 college roll no: 25EC13
Enter AC units consumed: 250
Enter rate per unit (Rs.): 8.5

--- HostelSense Report ---
Room: 3S10
Type: AC with Attached Bathroom

=== HOSTEL FEE ===
Student 1: P (Roll no: 25EC11)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Student 2: Q (Roll no: 25EC12)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Student 3: R (Roll no: 25EC13)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Total hostel fee from this room: Rs. 375000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 2125.00
P (Roll no: 25EC11) pays: Rs. 708.33
Q (Roll no: 25EC12) pays: Rs. 708.33
R (Roll no: 25EC13) pays: Rs. 708.33
Report saved successfully!
```

7.3 Contents of the saved file

After a successful run, the program writes the report into `HostelReport.txt` in the folder where it is run. The file holds exactly the report part of the console output, without the menu and the prompts. For example, after Run 1 the file contains:

```
--- HostelSense Report ---
Room: 3S12
Type: AC with Attached Bathroom

=== HOSTEL FEE ===
```

```
Student 1: D (Roll no: 25PE307)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Student 2: E (Roll no: 24ME300)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Student 3: F (Roll no: 25EL402)
   TOTAL HOSTEL FEE PAYABLE : Rs. 125000.00
Total hostel fee from this room: Rs. 375000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 300.00
D (Roll no: 25PE307) pays: Rs. 100.00
E (Roll no: 24ME300) pays: Rs. 100.00
F (Roll no: 25EL402) pays: Rs. 100.00
```

After Run 3 (non-AC room, 2S05) it contains:

```
--- HostelSense Report ---
Room: 2S05
Type: Non-Attached Bathroom, Non-AC

=== HOSTEL FEE ===
Student 1: G (Roll no: 25CE101)
   TOTAL HOSTEL FEE PAYABLE : Rs. 73000.00
Student 2: H (Roll no: 25CE102)
   TOTAL HOSTEL FEE PAYABLE : Rs. 73000.00
Student 3: I (Roll no: 25CE103)
   TOTAL HOSTEL FEE PAYABLE : Rs. 73000.00
Total hostel fee from this room: Rs. 219000.00

=== EXTRA ELECTRICITY BILL (paid separately) ===
Room electricity bill: Rs. 0.00
G (Roll no: 25CE101) pays: Rs. 0.00
H (Roll no: 25CE102) pays: Rs. 0.00
I (Roll no: 25CE103) pays: Rs. 0.00
```

If the program is run again, the old content is replaced by the new report.

8 Testing

The table below lists the test cases and what was expected and observed.

No.	Test	Expected result	Observed
1	Room type 3 (AC), units 300, rate 1	Fee 1,25,000 each; bill 300; share 100	As expected (Run 1)
2	Room type 1, three different roll nos.	Fee 95,000 each; bill 0	As expected (Run 2)
3	Room type 2, three different roll nos.	Fee 73,000 each; bill 0	As expected (Run 3)
4	Same roll number for students 1 and 2	Message shown, roll no. asked again	As expected (Run 4)
5	Room type 7	"Invalid choice!" and exit	As expected (Run 5)
6	AC room with rate −5	"Invalid electricity details!" and exit	As expected (Run 6)
7	AC room, 250 units at 8.5	Bill 2125.00; share 708.33	As expected (Run 7)
8	Report file after a run	`HostelReport.txt` has the same report as the screen	As expected

Table 6: Test cases

8.1 Verification of the amounts

Room	Type	Fee each	Room total	Elec.	Share
3S12	AC, attached bath	Rs. 1,25,000	Rs. 3,75,000	Rs. 300	Rs. 100
4S13	Non-AC, attached bath	Rs. 95,000	Rs. 2,85,000	Rs. 0	Rs. 0
2S05	Non-AC, no attached bath	Rs. 73,000	Rs. 2,19,000	Rs. 0	Rs. 0
3S10	AC, attached bath	Rs. 1,25,000	Rs. 3,75,000	Rs. 2,125	Rs. 708.33

Table 7: Results of the test runs

All of these match the hand calculation: $3 \times 125000 = 375000$, $3 \times 95000 = 285000$, $3 \times 73000 = 219000$, $300 \div 3 = 100$ and $2125 \div 3 = 708.33$.

9 Discussion

9.1 What worked well

- Passing the stream to `show()` kept the program short. The report format exists only once.
- Putting the electricity data in a derived class meant the base class did not have to carry fields that non-AC rooms never use.
- The duplicate check is a simple loop but it stops the most likely data entry error, which is entering the same student twice.

- Keeping the electricity bill as a separate section of the report makes it clear what is the fixed fee and what is the extra charge.

9.2 Points to be careful about

- Reading a name with spaces needs `getline`, and a leftover newline in the input buffer can make `getline` read an empty line. The `cin >> ws` part is there to avoid this.
- Money values are stored as `double`. This is fine for a small program like this, but a real billing system would normally use integers (paise) to avoid rounding errors. The share of Rs. 2125 among three students, for example, comes to 708.33 each, and three such shares add up to 2124.99, not 2125.
- The fee amounts are fixed numbers inside `main()`. If the hostel changes its fee, the code has to be edited and compiled again.

10 Limitations

These are the things the program does not do at the moment:

- It handles one room per run. To enter another room, the program has to be run again.
- The duplicate check only compares students inside the same room. The same roll number could still be entered again in a different run or a different room.
- The roll number is read as plain text and its format (for example `25EC407`) is not checked.
- Each run overwrites `HostelReport.txt`, so the report of the earlier room is lost unless the file is copied first.
- The number of students per room is fixed at three, and the fee amounts are written directly in the code.
- Only the units and rate are checked for invalid values. The room code and names are accepted as typed.

11 Future Scope

- Take many rooms in one run and check roll numbers across the whole hostel.
- Search for a student by roll number.
- Add fine or damage charges to the report.
- Append reports to the file instead of overwriting, or name each file after the room code.
- Read the fee amounts from a file so they can be changed without editing the code.
- Keep the data in a database, or build a GUI version.

12 Conclusion

HostelSense does what it was meant to do. The fee is chosen by the room type, a repeated roll number inside a room is rejected, and the AC electricity bill is split equally and kept apart from the hostel fee. The report is printed on screen and saved in a file using a single `show()` function.

The seven test runs, covering all three room types and the main error cases, gave the expected results, and the amounts agree with the hand calculations.

While making it, I got to use classes, inheritance, a loop-based validation check and file handling together in one program. The program is still basic, but the main idea works, and the limitations listed above show clearly what can be improved next.

References

1. Lab manual and course material, Object Oriented Programming (OOP 204ES), Birla Vishvakarma Mahavidyalaya.
2. E. Balagurusamy, *Object-Oriented Programming with C++*, 8th ed., McGraw Hill Education (India), New Delhi, 2021.
3. Online GDB, online C++ compiler: `https://www.onlinegdb.com`
4. C++ reference (`iostream`, `fstream`, `string`): `https://en.cppreference.com`

A Complete Source Code

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

class Room {
public:
    string roomCode;
    string names[3];
    string rollNo[3];

    void input() {
        cout << "Enter room code (e.g. 4S13): ";
        cin >> roomCode;

        for (int i = 0; i < 3; i++) {
            cout << "Enter student " << i + 1 << " name: ";
            getline(cin >> ws, names[i]);

            // unique college roll number
            while (true) {
                cout << "Enter student " << i + 1 << " college
                   roll no: ";
                cin >> rollNo[i];

                bool duplicate = false;
                for (int j = 0; j < i; j++) {
                    if (rollNo[j] == rollNo[i])
                        duplicate = true;
                }

                if (duplicate)
                    cout << "This roll no already lives in the
                       hostel! Enter again.\n";
                else
                    break;
            }
        }
    }
```

```
38
39      void show(ostream &out, string type, double bill, double
            fee) {
40          double share = bill / 3;
41
42          out.setf(ios::fixed);
43          out.precision(2);
44
45          out << "--- HostelSense Report ---\n";
46          out << "Room: " << roomCode << "\n";
47          out << "Type: " << type << "\n\n";
48
49          out << "=== HOSTEL FEE ===\n";
50          for (int i = 0; i < 3; i++) {
51              out << "Student " << i + 1 << ": " << names[i]
52                  << " (Roll no: " << rollNo[i] << ")\n";
53              out << "   TOTAL HOSTEL FEE PAYABLE : Rs. " << fee
                    << "\n";
54          }
55          out << "Total hostel fee from this room: Rs. " << fee *
                3 << "\n\n";
56
57          out << "=== EXTRA ELECTRICITY BILL (paid separately)
                ===\n";
58          out << "Room electricity bill: Rs. " << bill << "\n";
59          for (int i = 0; i < 3; i++) {
60              out << names[i] << " (Roll no: " << rollNo[i] << ")
                    pays: Rs. " << share << "\n";
61          }
62      }
63  };
64
65  class ACRoom : public Room {
66  public:
67      double units, rate;
68
69      ACRoom() {
70          units = 0;
71          rate = 0;
72      }
73
```

```
74      void inputAC() {
75          cout << "Enter AC units consumed: ";
76          cin >> units;
77          cout << "Enter rate per unit (Rs.): ";
78          cin >> rate;
79      }
80
81      double bill() {
82          return units * rate;
83      }
84  };
85
86  int main() {
87      int choice;
88      cout << "1. Attached bathroom, Non-AC     (Rs. 95,000)\n";
89      cout << "2. Non-attached bathroom, Non-AC (Rs. 73,000)\n";
90      cout << "3. AC with attached bathroom     (Rs. 1,25,000)\n"
            ;
91      cout << "Select room type: ";
92      cin >> choice;
93
94      if (cin.fail() || choice < 1 || choice > 3) {
95          cout << "Invalid choice!\n";
96          return 0;
97      }
98
99      Room r;
100     ACRoom c;
101     string type;
102     double bill = 0;
103     double fee = 0;
104
105     if (choice == 3) {
106         c.input();
107         c.inputAC();
108
109         if (cin.fail() || c.units < 0 || c.rate < 0) {
110             cout << "Invalid electricity details!\n";
111             return 0;
112         }
113         type = "AC with Attached Bathroom";
```

```
114             bill = c.bill();
115             fee = 125000;
116         }
117         else {
118             r.input();
119             if (choice == 1) {
120                 type = "Attached Bathroom, Non-AC";
121                 fee = 95000;
122             }
123             else {
124                 type = "Non-Attached Bathroom, Non-AC";
125                 fee = 73000;
126             }
127         }
128 
129         ofstream f("HostelReport.txt");
130         if (!f) {
131             cout << "Cannot create report!\n";
132             return 0;
133         }
134 
135         cout << "\n";
136         if (choice == 3) {
137             c.show(cout, type, bill, fee);
138             c.show(f, type, bill, fee);
139         }
140         else {
141             r.show(cout, type, bill, fee);
142             r.show(f, type, bill, fee);
143         }
144 
145         f.close();
146         cout << "Report saved successfully!\n";
147         return 0;
148     }
```